

NextGen Builder Initiative

Session 01 · Live Technical Webinar · Prepared for College Management Coordination

DATE & TIME	TOTAL DURATION	FORMAT	PANEL SIZE
18 Sep 2026, 9:50-12:10	2 hr 20 min	Fully Live Demo	4 Speakers

Session Overview

This session is a fully live, hands-on technical build-through of a real-world niche enterprise application, using an environment and codebase that is already built and running. Every segment demonstrates practical, industry-standard tooling in action — no slideware walkthroughs. Attendees will see live code, live infrastructure, and live deployment decisions exactly as they happen on real engineering teams, covering the full toolchain from requirement gathering through architecture, build, testing, and production operations.

Run-of-Show

#	SEGMENT	SPEAKER(S)	DURATION	CLOCK TIME
1	Introduction & Requirement Gathering	Manikandan	20 min	9:50 - 10:10 AM
2	Technology Stack Selection	Manikandan	20 min	10:10 - 10:30 AM
3	Database — Live Build	Mathavan / Manikandan	20 min	10:30 - 10:50 AM
4	Architecture — Live Walkthrough	Manikandan	10 min	10:50 - 11:00 AM
— BREAK — 5 minutes —				
5	Build & Testing — Live Code	Manikandan	20 min	11:05 - 11:25 AM
6	DevOps, Cloud & Observability — Live	Kalailvanan / Manikandan	20 min	11:25 - 11:45 AM
7	Closing — Career Guidance	Mannan	15 min	11:45 AM - 12:00 PM
8	Live Q&A — Pigeonhole	All Panelists	10 min	12:00 - 12:10 PM

TOTAL SESSION RUNTIME	2 hours 20 minutes (9:50 AM - 12:10 PM)
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Segment Detail — Live Coverage

1. Introduction & Requirement Gathering

Manikandan · 20 min

Live framing of a real niche enterprise application build. Walkthrough of how industry teams actually capture requirements — interviews, document study, exception-hunting — shown against the live requirement artifacts and user stories from the pre-built application.

2. Technology Stack Selection

Manikandan · 20 min

Live comparison of language and framework options against real industry decision criteria — hiring pool, ecosystem fit, long-term maintainability. The actual stack decision behind the running codebase is opened and reviewed live in the repository.

3. Database — Live Build

Mathavan / Manikandan · 20 min

Live walkthrough of the running schema — relational core, caching layer, and offline/edge data store. Live queries and migrations executed against the real environment.

4. Architecture — Live Walkthrough

Manikandan · 10 min

Live tour of the deployed system architecture — service boundaries, data flow, and the shape decision behind the running environment, shown directly from the live infrastructure rather than a diagram.

5. Build & Testing — Live Code

Manikandan · 20 min

Live pull request walkthrough on the real repository — branch, review, merge. The actual test suite is run live, including the concurrency and race-condition tests that matter most in production systems. AI-assisted code review shown in action.

6. DevOps, Cloud & Observability — Live

Kalailvanan / Manikandan · 20 min

Full live pipeline run: containerization with Docker, deployment to Kubernetes, the same workload shown running across AWS and Azure, and enterprise integration touch-points with SAP. Live CI/CD execution from lint through deploy, plus a live monitoring dashboard with a simulated incident and real-time rollback.

7. Closing — Career Guidance

Mannan · 15 min

Industry takeaways on what AI has changed in the engineering workflow versus what still requires human judgment. Career guidance for students and mid-level engineers attending the session.

8. Live Q&A — Pigeonhole

All Panelists · 10 min

Open floor — audience questions submitted via Pigeonhole answered live by the full panel.

Panelist Qualifications

Manikandan

Session Lead

Senior Full-Stack Architect and Engineering Lead, currently delivering an enterprise liquidity management and cash operations platform for a major banking client. Deep expertise in SWIFT messaging (MT940/942/101/103), MAS TRM compliance, and microservices architecture. Founder and Software Architect of Aadhirai Innovations, building ERP, billing, SaaS, and AI-assisted business intelligence solutions for enterprises and SMEs.

Mathavan

Data Architect

Independent consultant specializing in data architecture and database engineering, with hands-on experience designing and optimizing enterprise-grade database systems across client engagements.

Kalailvanan

Infrastructure & DevOps Lead

Vice President, Infrastructure and DevOps at DBS Bank, Singapore. Leads infrastructure strategy and DevOps practice for enterprise banking systems, with deep experience in production-grade deployment, cloud operations, and platform reliability at scale.

Mannan

Product Owner

Product Owner at Volvo, bringing an industry product-management perspective on translating business requirements into engineering roadmaps and guiding delivery from concept through release.

Note to organizers: The full 2 hour 20 minute runtime (9:50 AM - 12:10 PM) includes one 5-minute break after the Architecture segment. Kindly plan hall/lab availability and refreshment arrangements accordingly.